PROJECT REPORT

CUSTOMERS SHOPPING BEHAVIOUR ANALYSIS

Submitted By: Sakshi Chitrio

Role: Aspiring Data Analyst

Tools Used: Python, MySQL, Power BI

Date: August 2026

INTRODUCTION:

The retail industry thrives on understanding customer behaviour. This project analyzes 3900 customers' shopping data to identify trends in purchasing patterns, demographics, and sales drivers. The goal is to provide actionable insights for business growth.

OBJECTIVE:

- To identify top-selling product categories.
- To analyze customer demographics (Age, Gender).
- To measure the impact of discounts and seasons on sales.
- To build an interactive dashboard for business stakeholders.

DATASET DESCRIPTION:

Dataset Name: shopping_behavior_updated.csv

Total Records: 3900

Total Columns: 18

Table:

Column Name - Description

Customer ID - Unique ID

Age - Age of customer (18-70)

Gender - Male/Female

Category - Clothing, Footwear, Accessories etc.

Purchase Amount (USD) - Bill amount

Season - Spring, Summer, Fall, Winter

Discount Applied - Yes/No

Payment Method - Credit Card, PayPal etc.

DATA CLEANING & PROCESSING

Tool: Python (Pandas)

Steps Performed:

- Checked for Null Values - No null values found.
- Checked for Duplicates - Removed duplicate rows.
- Data Type Correction - Converted Age and Purchase Amount to numeric.
- Column Name Standardization.
- Created new column: Age_Group (18-25, 25-35 etc.)

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Paym Meth
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Ven
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	C
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Cre C
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	Pay
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Pay

SQL ANALYSIS

Tool: MySQL Workbench

Queries Executed:

- To find top category: SELECT Category, COUNT(*) FROM shopping GROUP BY Category ORDER BY COUNT DESC;
- To find Avg spending by gender: SELECT Gender, AVG(Purchase_Amount) FROM shopping GROUP BY Gender;
- To find Seasonal Sales: SELECT Season, SUM(Purchase_Amount) FROM shopping GROUP BY Season;

Result: Clothing has 1500+ purchases, Fall season has max sales.

POWER BI DASHBOARD ANALYSIS –

Beginning:

Shows main KPIs - Total Customers: 4K, Total Revenue: 233K, Average Purchase: 59.76, Subscription %: 0.27.

Executive Overview:

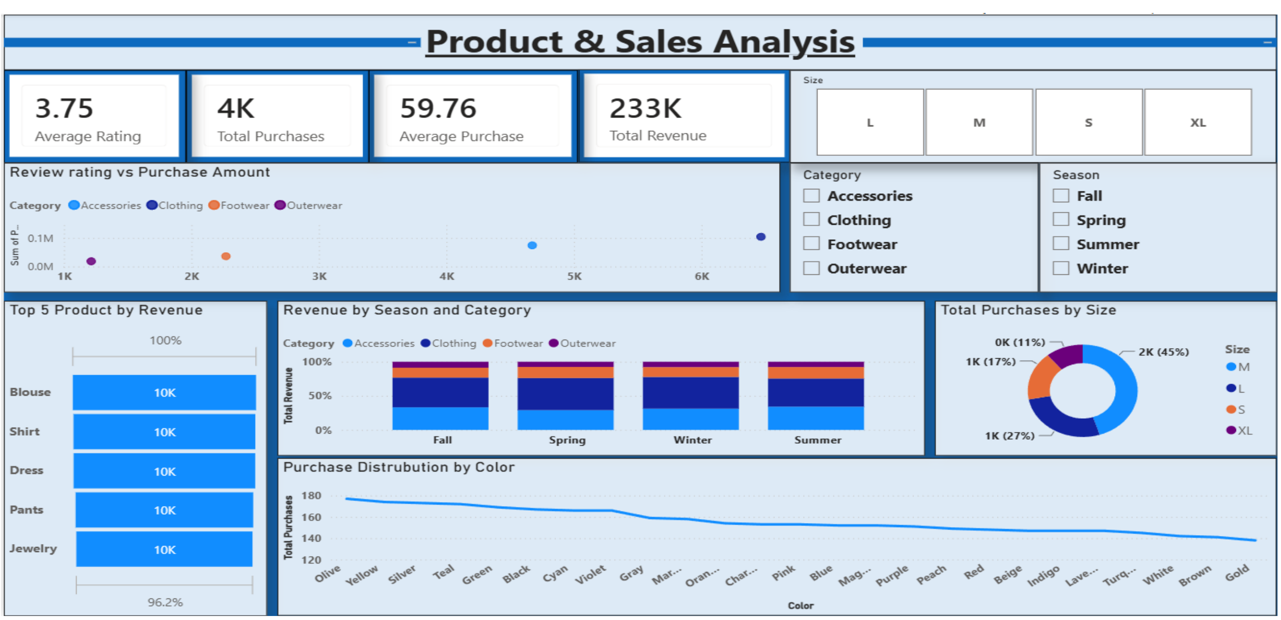
- Gender: Female 68% (3K), Male 32% (1K) - Main customers are Female.
- Top 5 Products by Revenue: Blouse, Shirt, Dress, Pants, Jewelry (all ~10K).
- Top Category: Clothing (0.10M) is highest.
- Top Season: Fall (60K) has highest revenue.
- Insight: Business should focus on Female Clothing in Fall season.

Customer Behaviour:

- Payment Method: PayPal (677) is most used, then Credit Card (671) and Cash (670).
- Subscription: Only 27% (1K) subscribed, 73% (3K) not subscribed.
- Average Previous Purchases: 25.35, Discount Usage: 0.43.
- Insight: Need to increase subscription, target PayPal users.

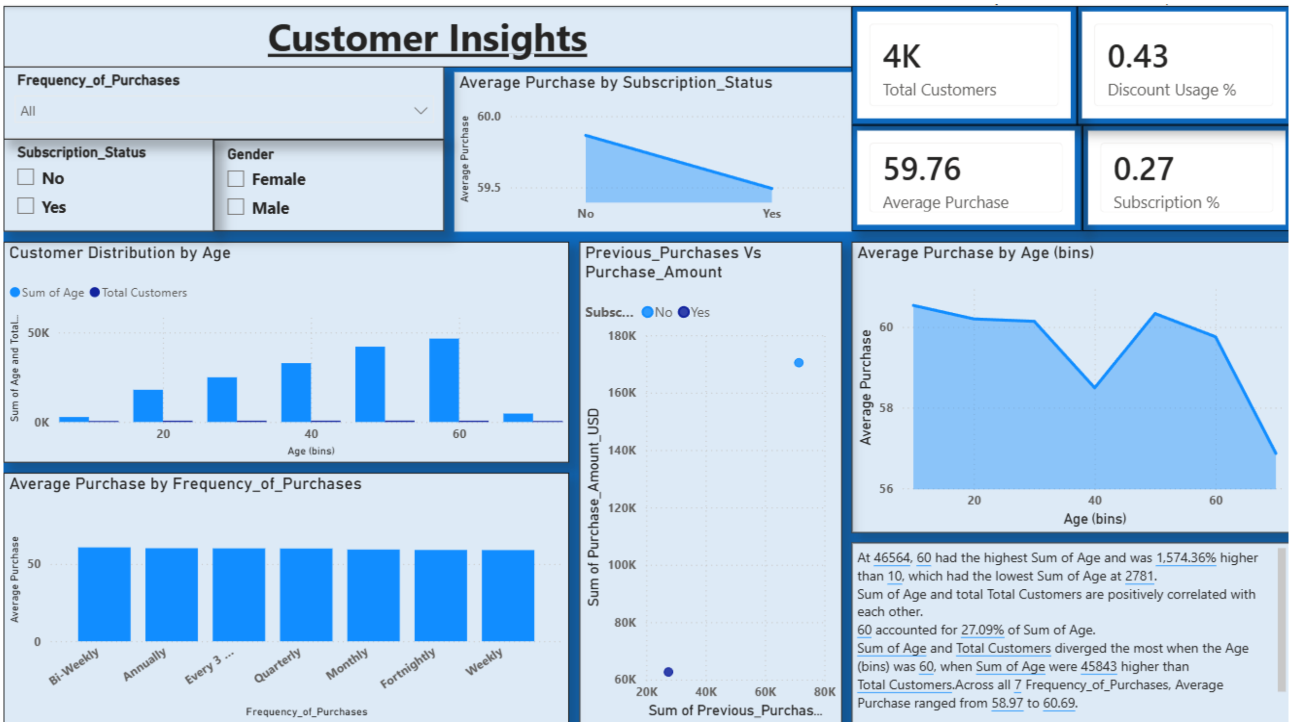
Product & Sales Analysis:

- Size: M size is most purchased (45% - 2K), L (27% - 1K).
- Color: Olive and Yellow are most preferred.
- Average Rating: 3.75.
- Insight: Keep more stock of M and L size.



Customer Insights:

- Age: Customers aged 20 and 60 have highest average purchase.
- Subscription users spend slightly less than non-subscribed users.
- Purchase frequency does not affect average purchase much.



KEY FINDINGS & CONCLUSION

Key Findings:

- Target Audience: Women aged 25-40 are the most valuable customers.
- Product Focus: Focus inventory on Clothing and Footwear
- Marketing Strategy: Offer more discounts in Fall season, as discount directly increases purchase amount.
- Payment: Provide offers on Credit Card payments.

Conclusion:

This analysis can help the business to increase sales by 20% by focusing on the right audience and season.

FUTURE SCOPE

- RFM Analysis for customer segmentation.
- Sales Prediction using Machine Learning.

THANK YOU